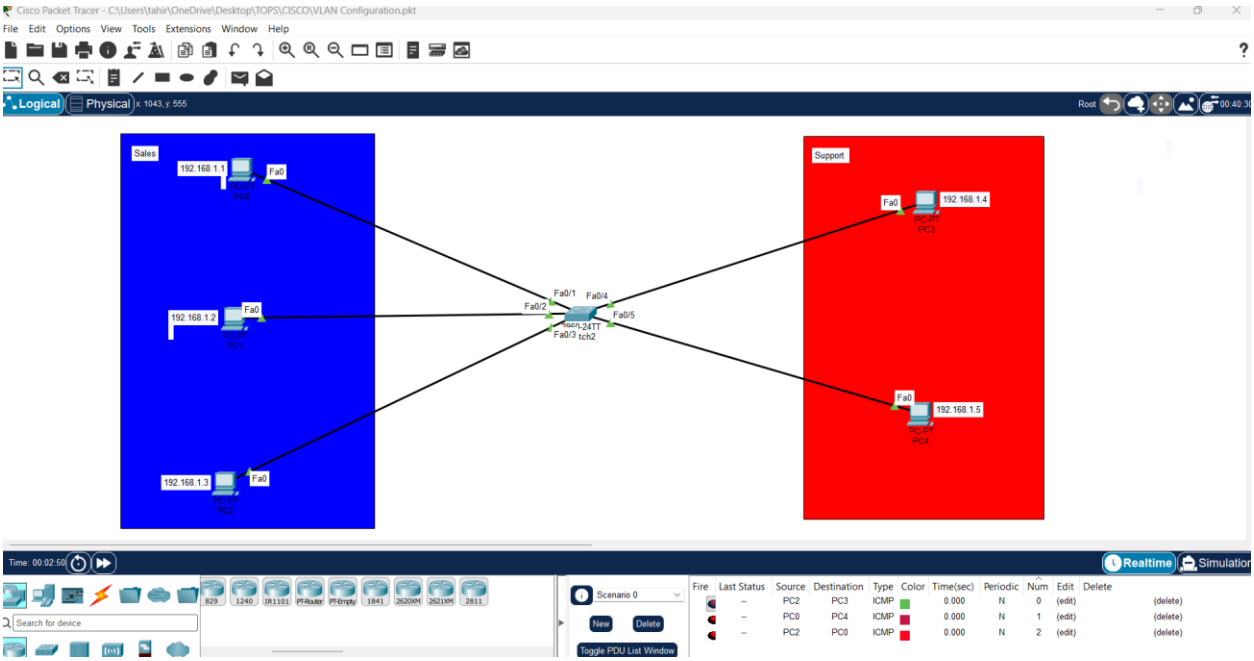
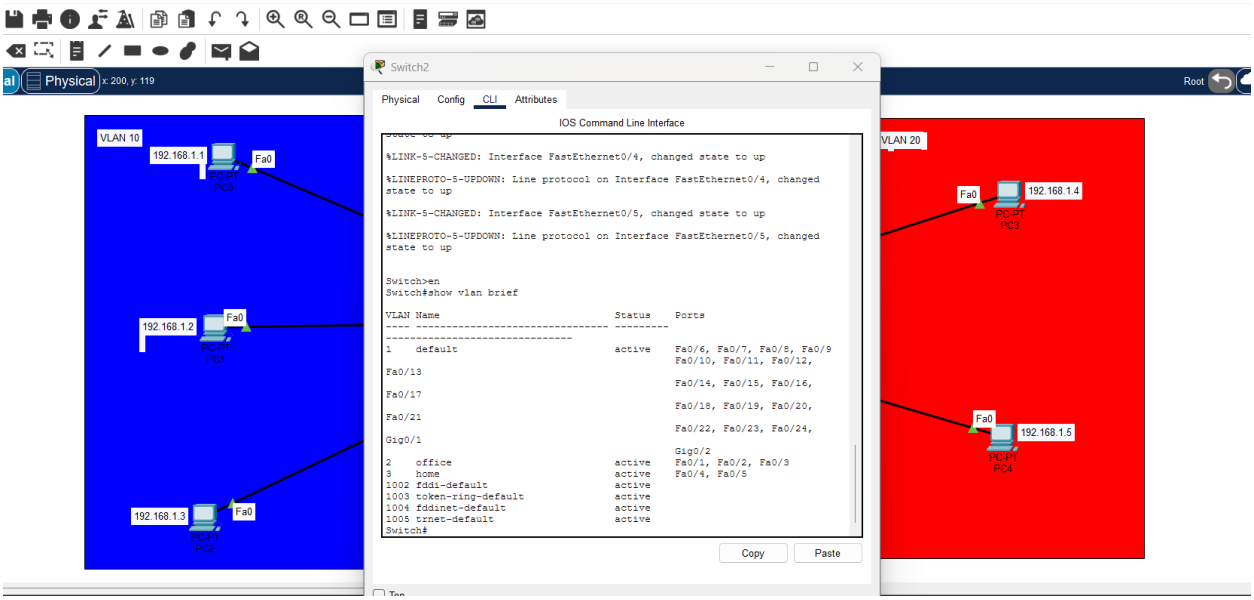
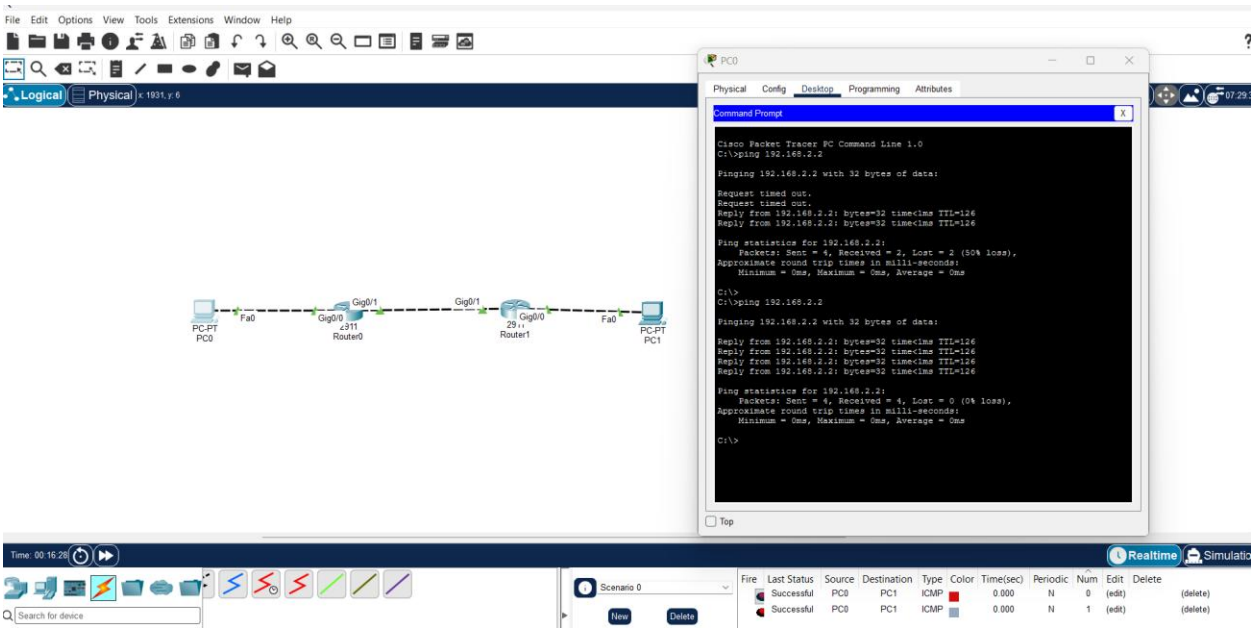
1. **Log in to Cisco Packet Tracer or any network simulation tool, add a switch, and create two VLANs named 'Sales' and 'Support' with VLAN IDs 10 and 20.**



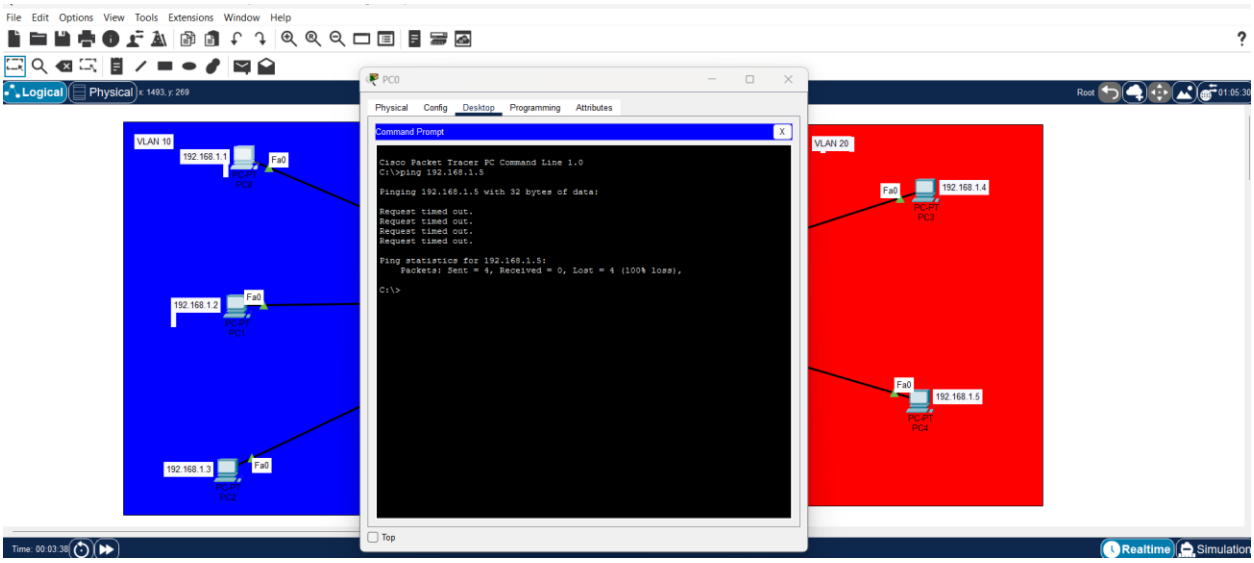
2. **Assign three different switch ports to VLAN 10 (Sales) and two ports to VLAN 20 (Support), then verify the VLAN membership using the show vlan command. Hint: Use the interface range command to select multiple ports efficiently.**



3. **Connect two routers using a serial or Ethernet link in your simulation tool, assign IP addresses to their interfaces, and configure a static route on Router 1 to reach the LAN behind Router 2.**



4. **Simulate a scenario where a device in the 'Sales' VLAN tries to ping a device in the 'Support' VLAN and explain why the ping fails, then suggest what configuration would allow inter-VLAN communication. Hint: Think about router-on-a-stick or Layer 3 switch concepts.**



❖ Why does the ping fail?

VLAN 10 (Sales) and VLAN 20 (Support) are separate networks. A Layer 2 switch cannot send data directly between them.

❖ **How can we fix it?**

We can use Router-on-a-Stick by connecting a router to the switch and configuring sub-interfaces for both VLANs. Alternatively, we can use a Layer 3 switch with routing enabled.

- **In simple words:** Devices in different VLANs cannot communicate directly. We need a router or Layer 3 switch to allow communication between them.